

Rhilo Sotto
2/5/2025
COMPE 470L

Lab 3 Report

Introduction

In this lab we design a hexadecimal counter with a 7 Segment Decoder that starts at 'F', counts down to '0' and repeats. We implement it in Verilog alongside a constraint file that interfaces and maps the Verilog module with the physical ports on the Basys3 FPGA. I used the code from fpga4student.com to get the Basys3 constraint and clock files, and implemented the hex signals for 'A' through 'F'.

Verilog Code

Seven Segment Display Module

```
// fpga4student.com: FPGA projects, Verilog projects, VHDL projects
// FPGA tutorial: seven-segment LED display controller on Basys 3 FPGA
module Seven_segment_LED_Display_Controller(
    input clock_100Mhz, // 100 Mhz clock source on Basys 3 FPGA
    input reset, // reset signal
    output reg [3:0] Anode_Activate, // anode signals of the 7-segment LED display
    output reg [6:0] LED_out// cathode patterns of the 7-segment LED display
);
    reg [26:0] one_second_counter; // counter for generating 1 second clock enable
    wire one_second_enable;// one second enable for counting numbers
    reg [15:0] displayed_number; // counting number to be displayed
    reg [3:0] LED_BCD;

    always @(posedge clock_100Mhz or posedge reset)
    begin
        if(reset == 1)
            one_second_counter <= 0;
        else begin
            if(one_second_counter >= 999999999)
                one_second_counter <= 0;
            else
                one_second_counter <= one_second_counter + 1;
        end
    end
end
```

```

assign one_second_enable = (one_second_counter == 99999999) ? 1 : 0;

always @(posedge clock_100Mhz or posedge reset)
begin
    if(reset==1)
        displayed_number <= 4'b1111; // start at 'F'
    else if(one_second_enable == 1)
        displayed_number <= (displayed_number == 0) ? 4'b1111 : (displayed_number - 1);
    end

// decoder to generate anode signals
always @(*) begin
    Anode_Activate = 4'b1110;
    LED_BCD = displayed_number;
end

// Cathode patterns of the 7-segment LED display
always @(*)
begin
    case(LED_BCD)
        4'b0000: LED_out = 7'b0000001; // "0"
        4'b0001: LED_out = 7'b1001111; // "1"
        4'b0010: LED_out = 7'b0010010; // "2"
        4'b0011: LED_out = 7'b0000110; // "3"
        4'b0100: LED_out = 7'b1001100; // "4"
        4'b0101: LED_out = 7'b0100100; // "5"
        4'b0110: LED_out = 7'b0100000; // "6"
        4'b0111: LED_out = 7'b0001111; // "7"
        4'b1000: LED_out = 7'b0000000; // "8"
        4'b1001: LED_out = 7'b0000100; // "9"

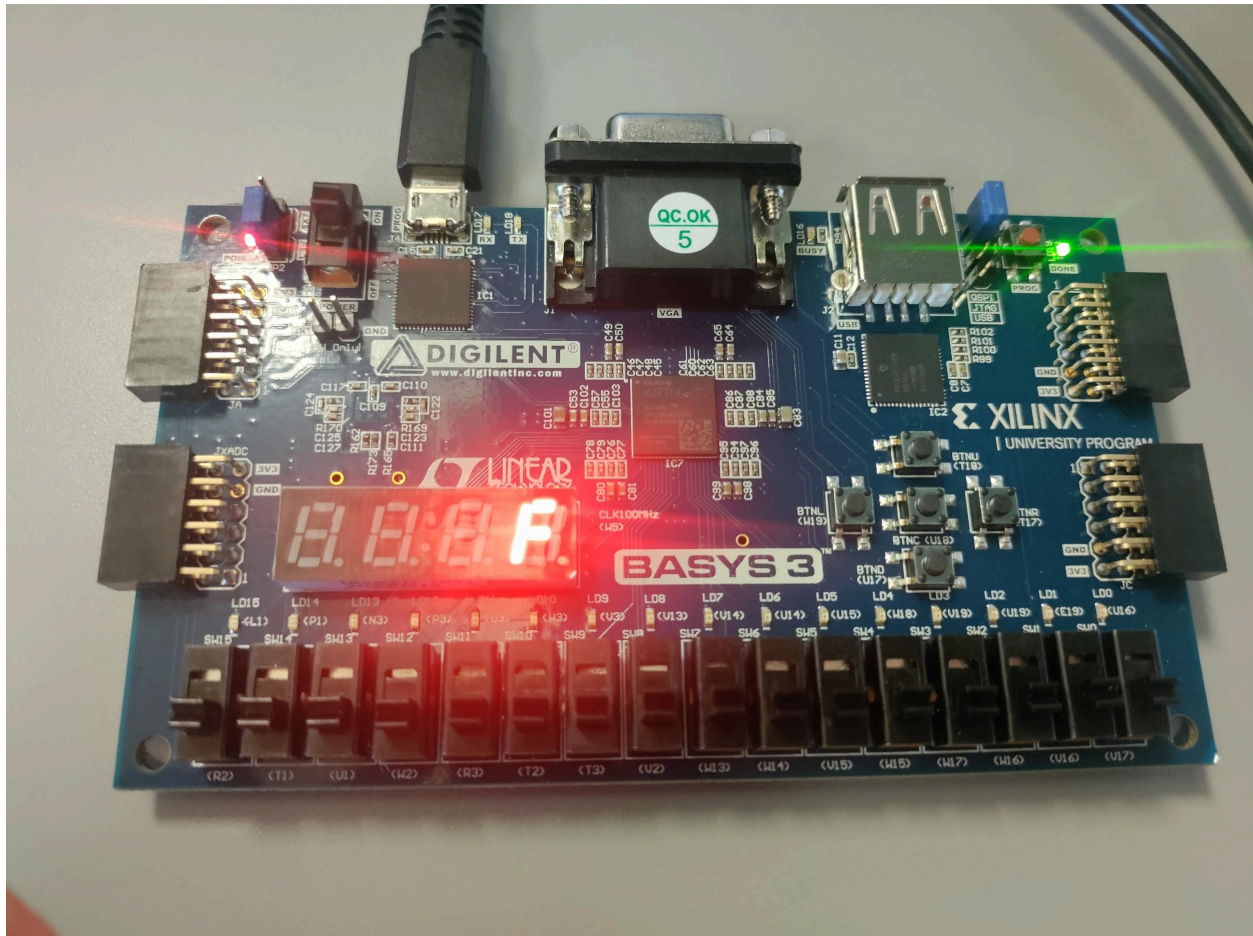
        4'b1010: LED_out = 7'b0001000; // "A"
        4'b1011: LED_out = 7'b1100000; // "B"
        4'b1100: LED_out = 7'b0110001; // "C"
        4'b1101: LED_out = 7'b1000010; // "D"
        4'b1110: LED_out = 7'b0110000; // "E"
        4'b1111: LED_out = 7'b0111000; // "F"
        default: LED_out = 7'b0000001; // "0"
    endcase
end
endmodule

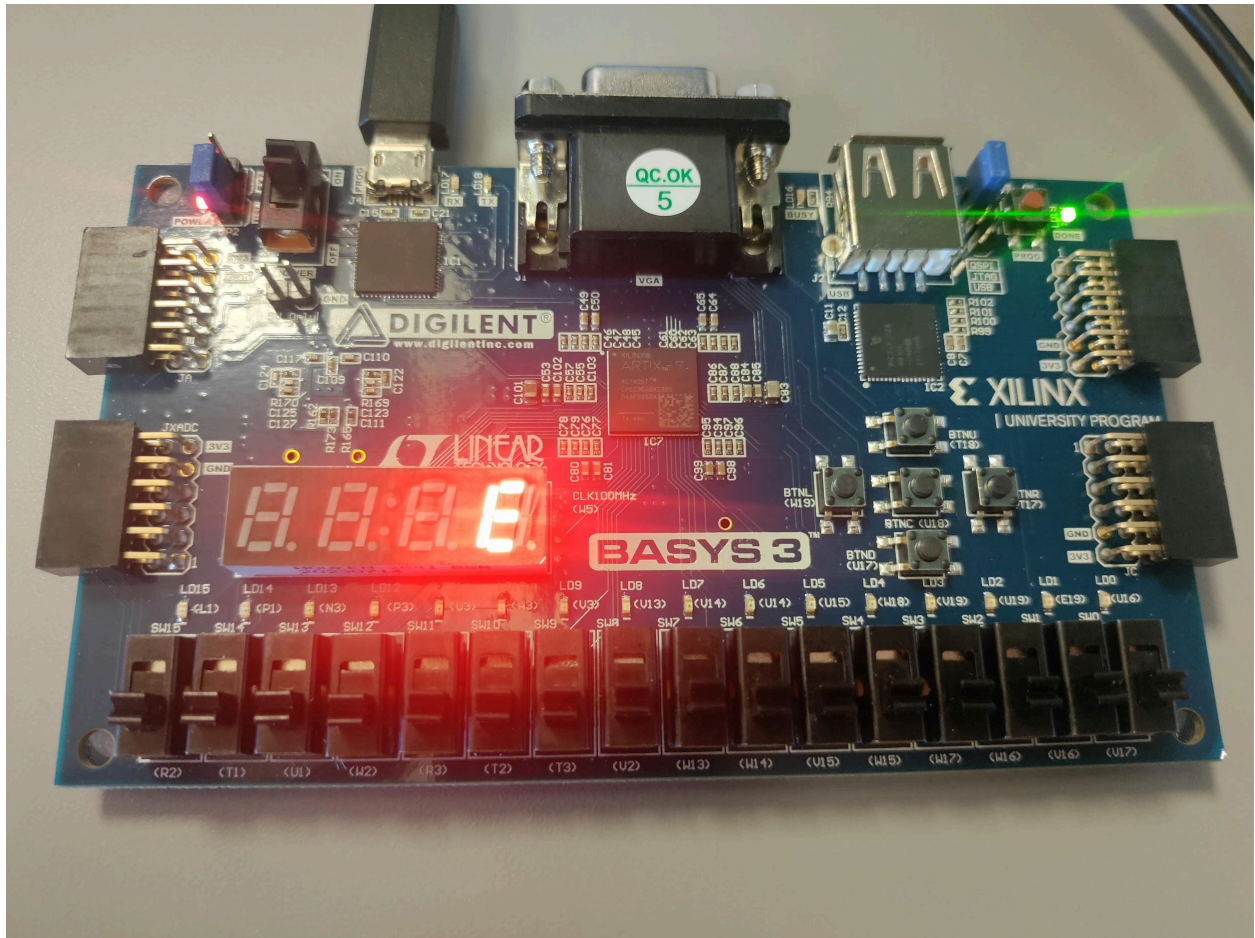
```

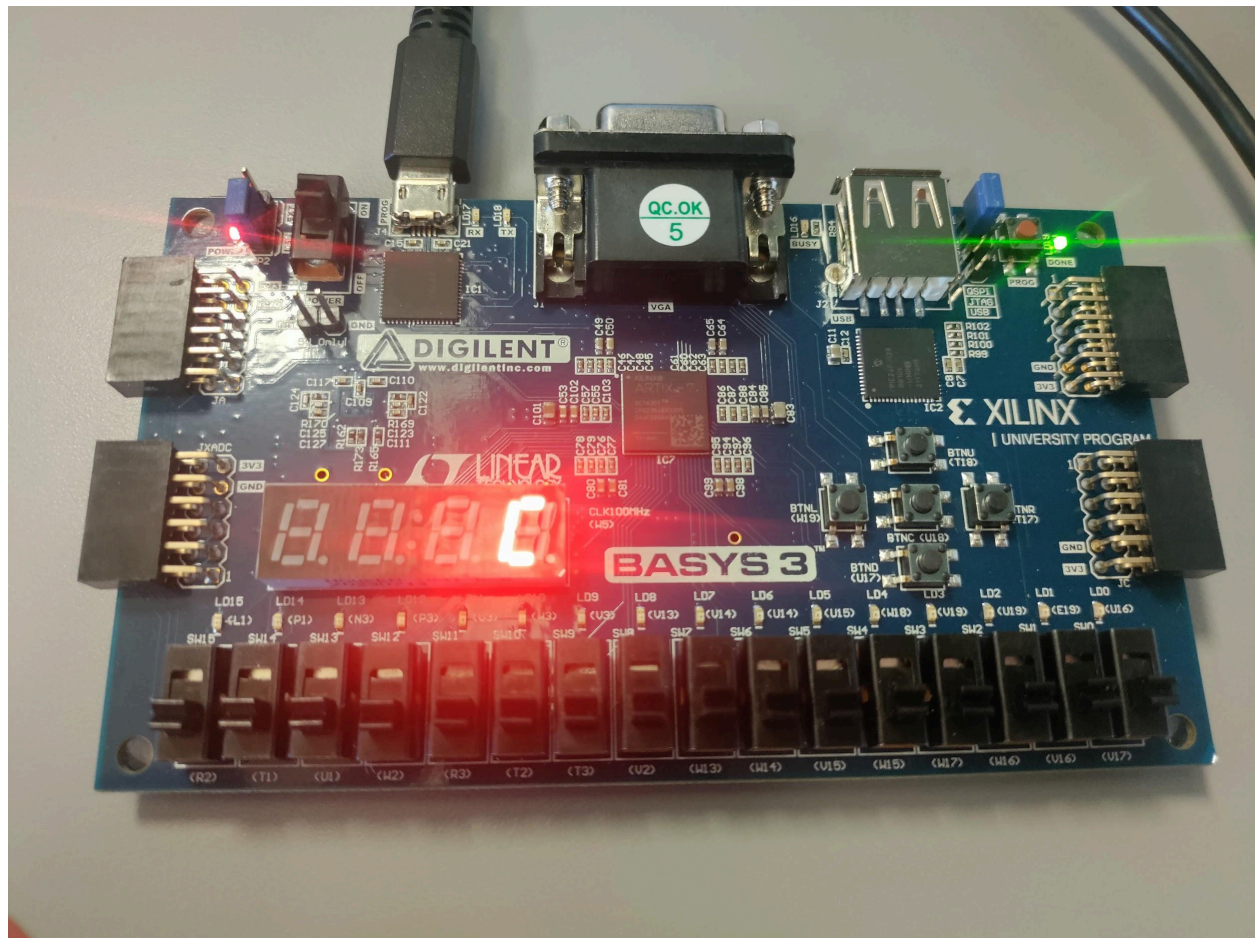
Constraint File

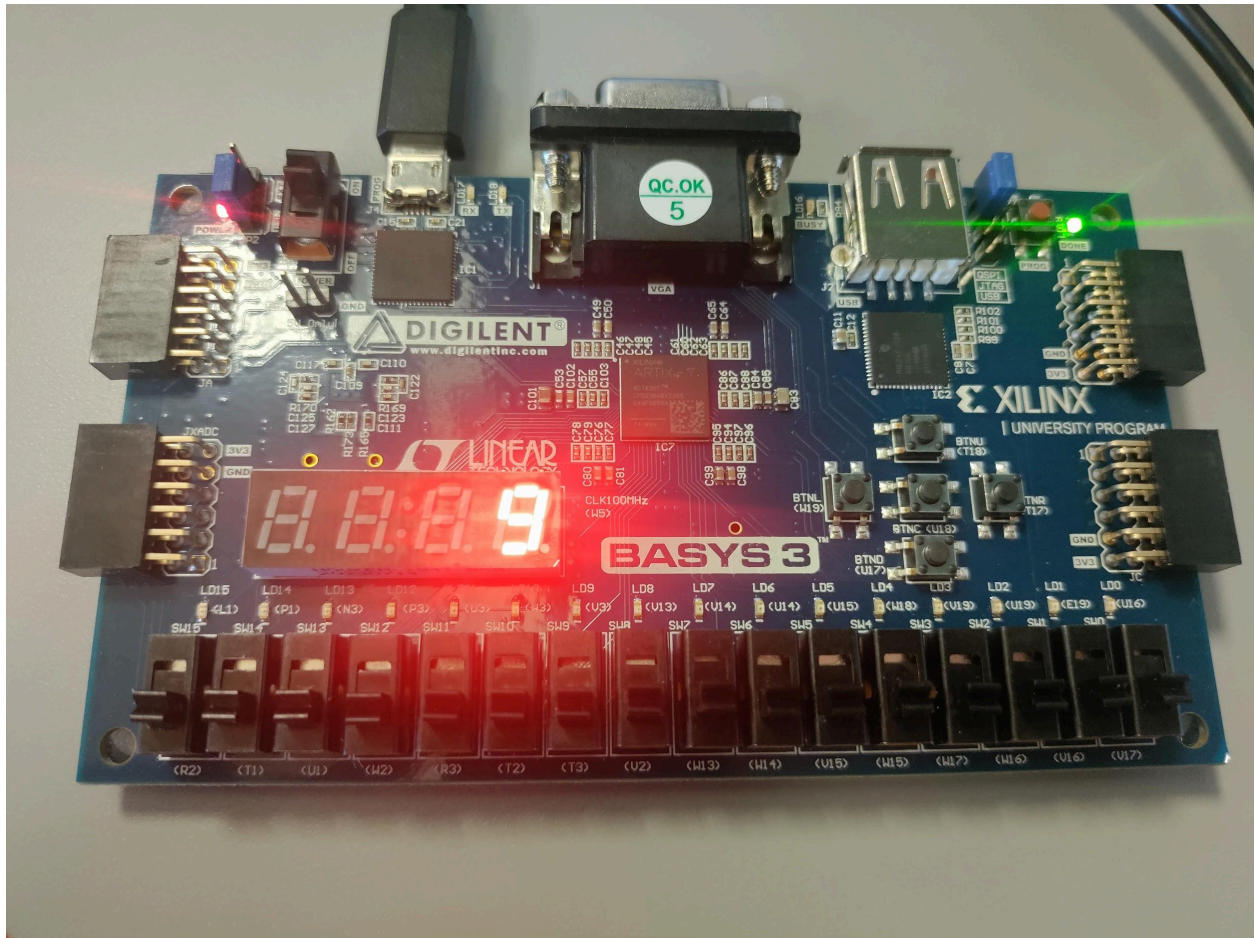
```
# Clock signal
set_property PACKAGE_PIN W5 [get_ports clock_100Mhz]
set_property IOSTANDARD LVCMOS33 [get_ports clock_100Mhz]
set_property PACKAGE_PIN R2 [get_ports reset]
set_property IOSTANDARD LVCMOS33 [get_ports reset]
#seven-segment LED display
set_property PACKAGE_PIN W7 [get_ports {LED_out[6]}]
set_property IOSTANDARD LVCMOS33 [get_ports {LED_out[6]}]
set_property PACKAGE_PIN W6 [get_ports {LED_out[5]}]
set_property IOSTANDARD LVCMOS33 [get_ports {LED_out[5]}]
set_property PACKAGE_PIN U8 [get_ports {LED_out[4]}]
set_property IOSTANDARD LVCMOS33 [get_ports {LED_out[4]}]
set_property PACKAGE_PIN V8 [get_ports {LED_out[3]}]
set_property IOSTANDARD LVCMOS33 [get_ports {LED_out[3]}]
set_property PACKAGE_PIN U5 [get_ports {LED_out[2]}]
set_property IOSTANDARD LVCMOS33 [get_ports {LED_out[2]}]
set_property PACKAGE_PIN V5 [get_ports {LED_out[1]}]
set_property IOSTANDARD LVCMOS33 [get_ports {LED_out[1]}]
set_property PACKAGE_PIN U7 [get_ports {LED_out[0]}]
set_property IOSTANDARD LVCMOS33 [get_ports {LED_out[0]}]
set_property PACKAGE_PIN U2 [get_ports {Anode_Activate[0]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Anode_Activate[0]}]
set_property PACKAGE_PIN U4 [get_ports {Anode_Activate[1]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Anode_Activate[1]}]
set_property PACKAGE_PIN V4 [get_ports {Anode_Activate[2]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Anode_Activate[2]}]
set_property PACKAGE_PIN W4 [get_ports {Anode_Activate[3]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Anode_Activate[3]}]
```

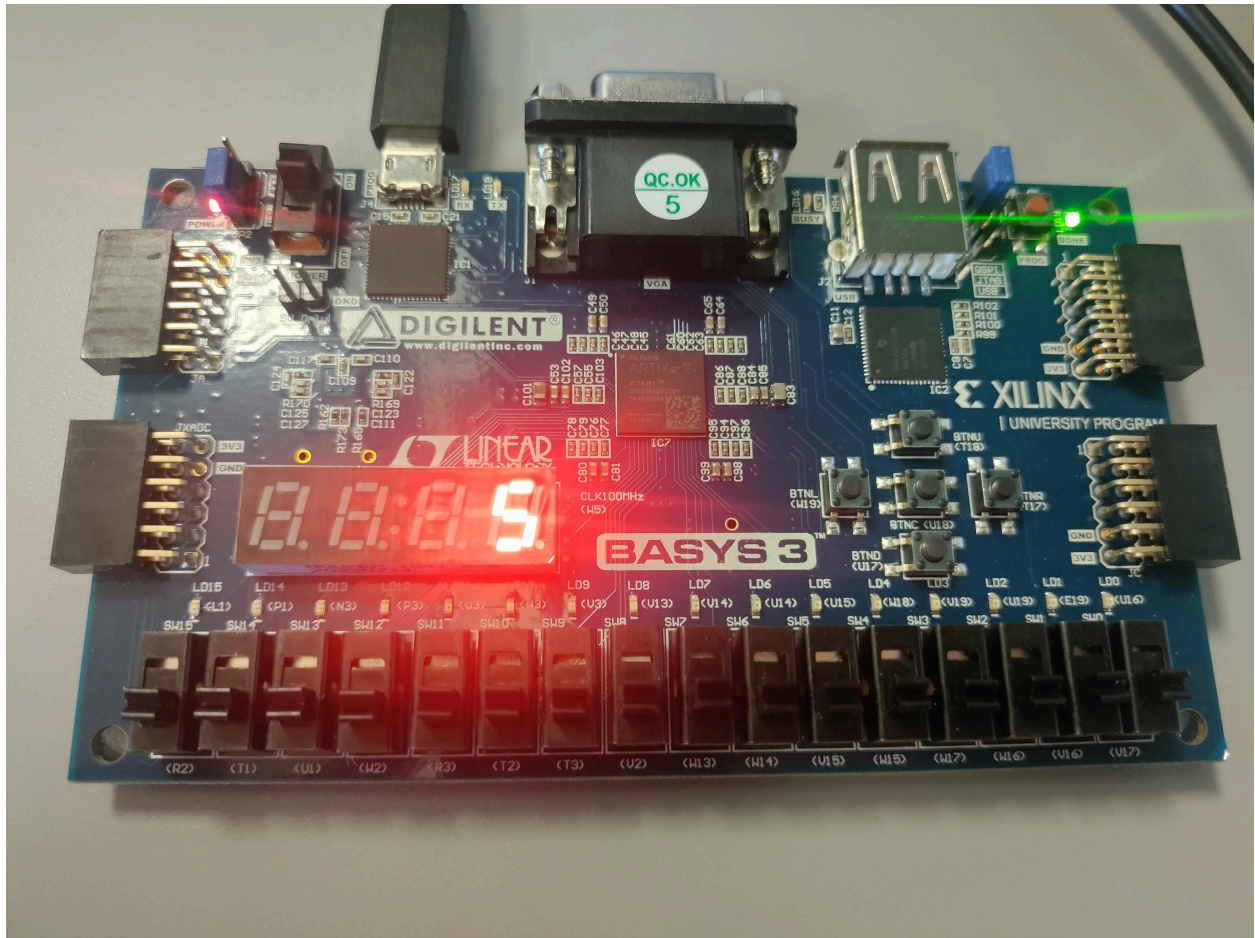
Results

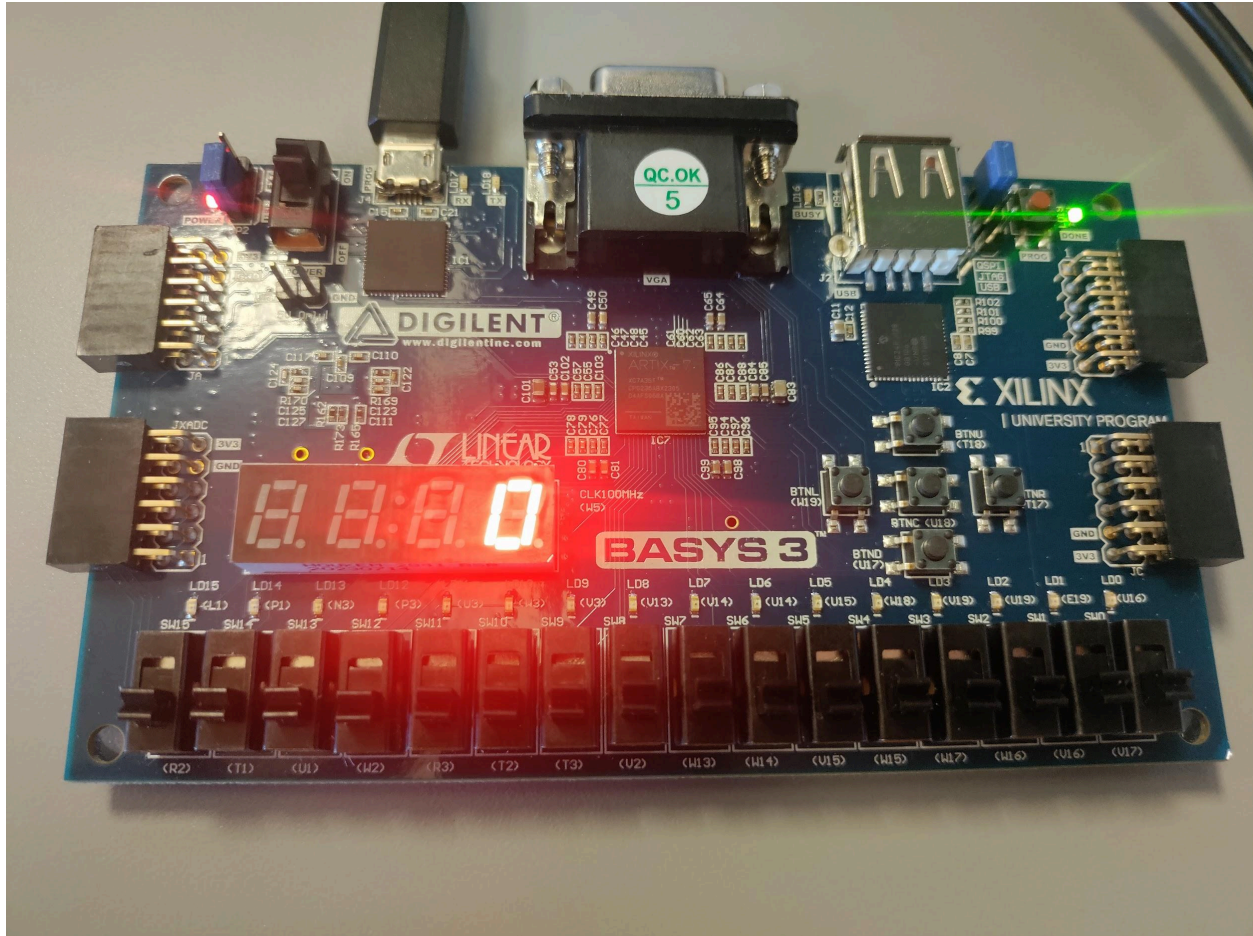












Conclusion

In this lab we programmed a Basys3 FPGA in Verilog and successfully implemented a hexadecimal counter with a 7 Segment Decoder using the onboard LEDs, we toggled the anodes to make it so only one of the LEDs was active, and the cathode signals to light up the segment needed to display the digit. We also learned how to interface the physical pins and ports in Verilog using constraint files.